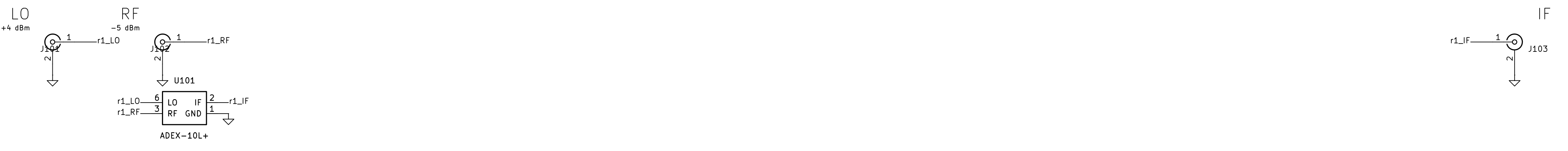
r1 diode mixer, ADEX-10L+

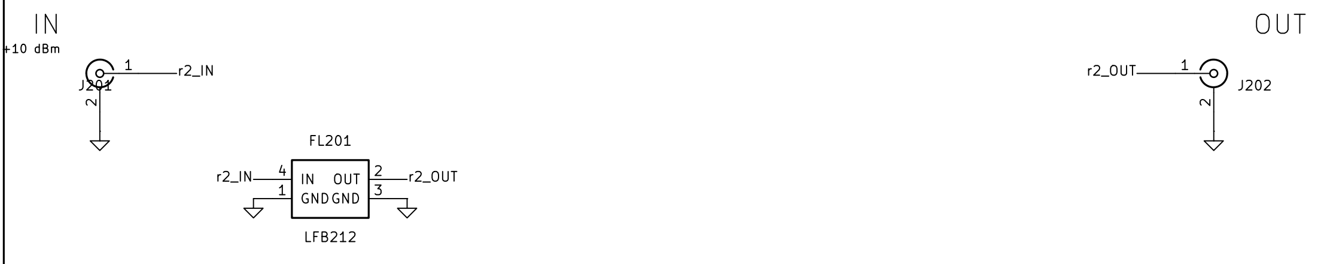
cells (0,0), (1,0)



EXPECT: about 7 dB of conversion loss from 10 MHz to 1 GHz with +4 dBm of LO (r10 at its top power), IF from DC to 800 MHz, 60 dB L−R isolation
Mini−Circuits ADEX−10L+: two wideband baluns and a ring of four Schottky diodes (drawn beside it on the silk). The LO (+4 dBm, +1 to +7) turns the diodes on in pairs, so the RF is multiplied by +−1 at the LO rate: the IF is RF x square wave, the sum and difference plus odd LO harmonics. Passive and DC−coupled at IF (a phase detector when RF and LO share a frequency); 1 dB compression at +1 dBm of RF, IP3 +16 dBm. Compare r7/r8 (ADL5350) and AMOD−A's a9 (MC1496).

r2 2.45 GHz filter

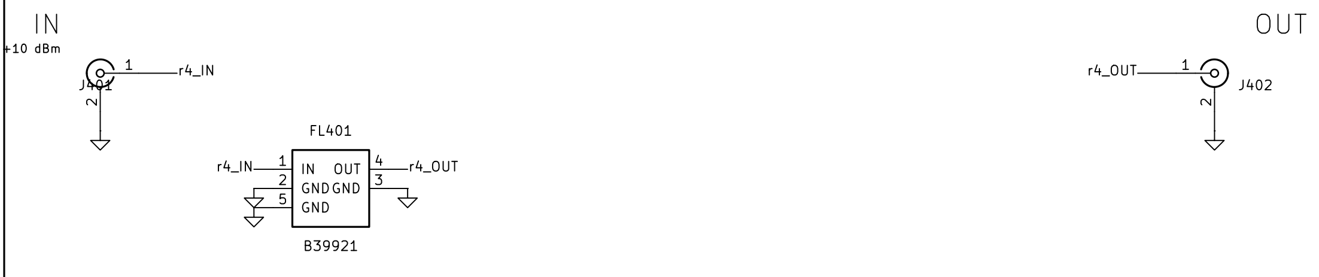
cell (2,0)



EXPECT: 1.7 dB of loss from 2.4 to 2.5 GHz, 30 dB or more below 2.2 GHz and above 2.7 GHz (the datasheet); 50 ohm in and out
Murata LFB212G45SG8A127, a 2012−size LTCC band−pass. The line to each jack is 50 ohm coplanar waveguide (0.29 mm wide, 0.25 to the ground pour) with a via fence; the jack's centre pin has no plane under it (a 2.2 mm void).
Sweep it with r10 or r13 into a power meter, or with a DAC harmonic comb and the ADC.

r4 915 MHz filter

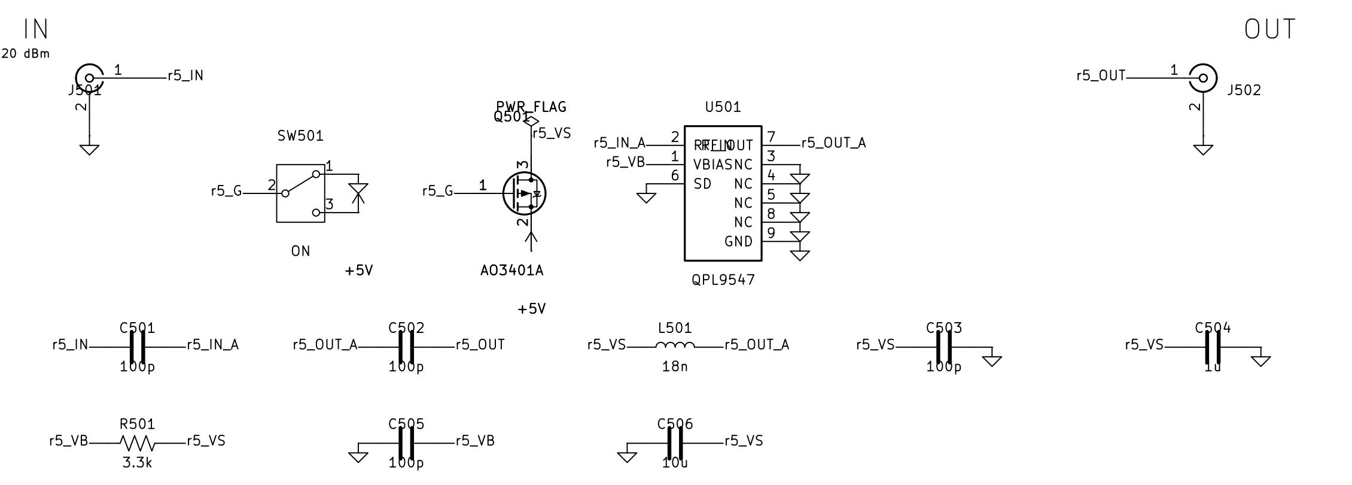
cell (3,0)



EXPECT: 1.1 dB of loss from 902 to 928 MHz, 40 dB or more 30 MHz outside the band (the datasheet); 50 ohm in and out
RF360 B39921B2672P810: a surface−acoustic−wave filter for the 915 MHz ISM band, 26 MHz wide, in a 1411 package; its skirts are much steeper than the LTCC's. Lines, fences and launch voids as on r2.

r5 LNA, QPL9547

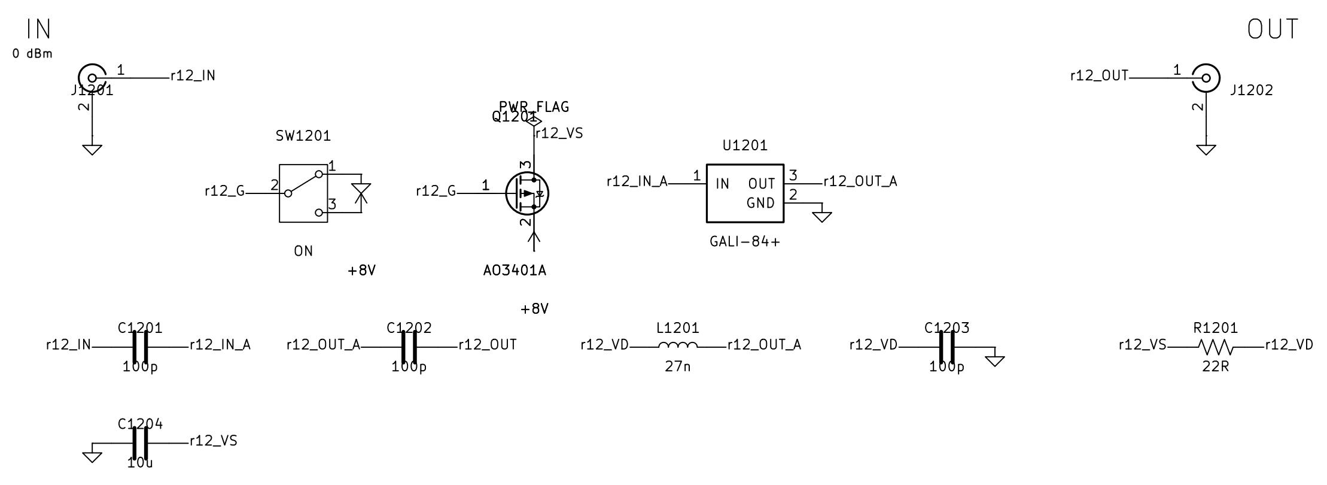
cell (0,1)



EXPECT: 19.5 dB of gain and a 0.3−0.5 dB noise figure from 0.5 to 3 GHz (datasheet: 0.3 dB at 1.9 GHz, OIP3 +39 dBm); 65 mA from 5 V
Qorvo QPL9547: a GaAs pHEMT LNA, internally matched to 50 ohm, 0.1−6 GHz. It sets the noise figure of every receiver here, so put it first (before the filter) unless a strong neighbour overloads it; +39 dBm OIP3 makes that rare.
The 18 nH wire−wound choke (Qorvo's value) feeds 5 V into the output pin; 3.3 k from 5 V to Vbias sets 65 mA.
Never cable a transmitter into it without 30 dB of attenuation (22 dBm absolute maximum at the input).

r12 gain block, GALI−84+

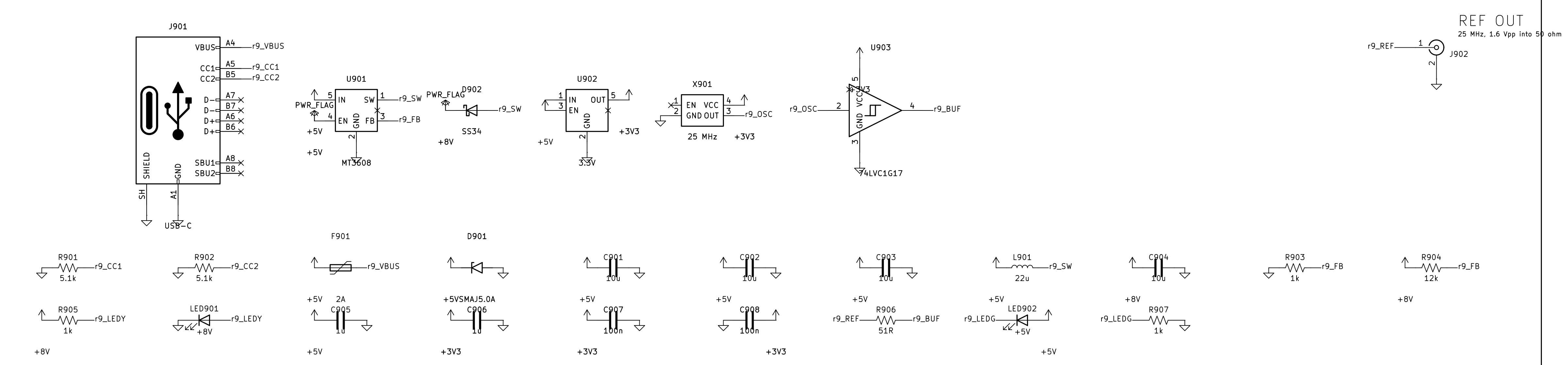
cell (1,1)



EXPECT: 20 dB of gain from 50 MHz to 2.5 GHz (datasheet: 21.9 dB at 1 GHz, 1 dB compression at +20 dBm out); 97 mA from 7.8 V
Mini−Circuits GALI−84+: a Darlington MMIC that sets its own current; the 22 ohm from the 7.8 V rail drops to its 5.8 V device voltage (0.2 W in the 2512). The 27 nH choke (415 ohm at 2.45 GHz, 155 at 915 MHz) feeds the output pin; 100 pF blocks keep the DC off the jacks. Keep the input under 0 dBm for linear gain. The slide switch and P−FET gate the tile's 7.8 V as on AMOD−A.

r9 power, USB−C 5 V in: +5, +7.8 V; 25 MHz reference

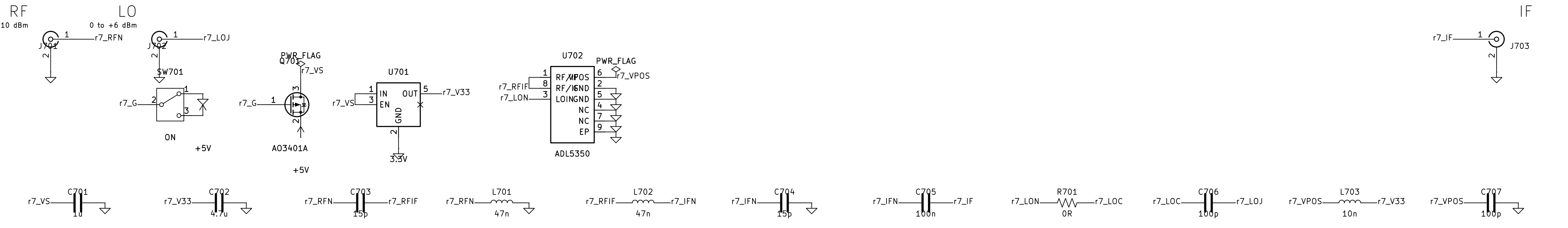
cells (2,1), (3,1)



EXPECT: +5 V from the USB−C (up to 1.5 A), +7.8 V at 250 mA for the gain blocks (yellow LED), a 25 MHz square wave on REF OUT
USB−C power only: 5.1 k on each CC asks any source for 5 V; a 2 A PTC, a 5 V TVS and 20 uF. The MT3608 boosts to 7.8 V (0.6 V x 13 from 12 k / 1 k) for the GALI−84+ tile, r12 (100 mA); the +8V rail is a strip under row 1 that their headers reach with a via. The TPS7A2033 makes 3.3 V for the TC32H4 TCXO (+−2 ppm) and the 74LVC1G17 that drives REF OUT (51 ohm: 1.6 Vpp into 50 ohm, 3.3 Vpp into a synthesizer's unterminated REF IN; r10's REF IN passes it to r13).
The TCXO's pad 1 (enable or no connection on the drawing) floats, as CostasLink's review settled.

r7 TX mixer, ADL5350

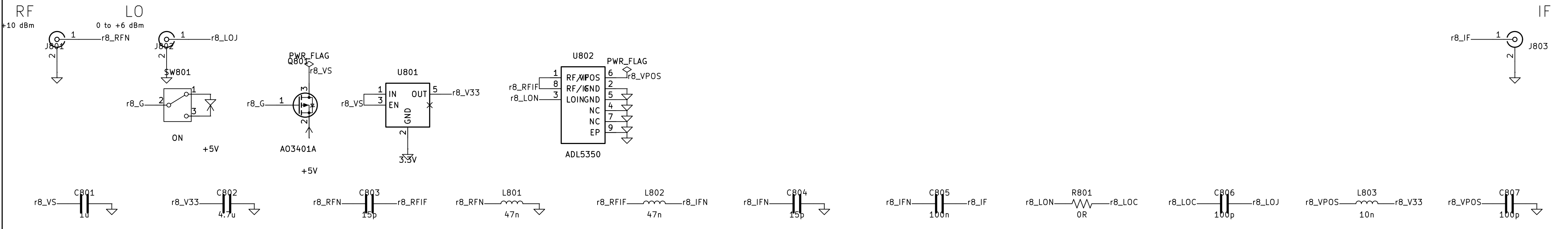
cells (0,2), (1,2)



EXPECT: RF x LO to IF (or IF x LO to RF) with 7−9 dB of loss from 400 MHz to 2.5 GHz and an IF of 0.1−30 MHz; +4 dBm of LO is enough
ADL5350: a GaAs FET switch mixer with its own LO buffer; the RF and IF pins are one node, split by a crossover (series 15 pF + shunt 47 nH toward RF, series 47 nH + shunt 15 pF toward IF, corner 200 MHz) in place of the datasheet's band−tuned tanks: a few dB more loss, every band at once. The LO's series element is 0 ohm (fit the datasheet's 3−8 nH for one band); the LO bias choke is 10 nH (the datasheet: 24 nH at 780 MHz, 2 nH at 2 GHz).
Passive, so the same tile up−converts a DAC tone (IF in, RF out) or down−converts an antenna's signal (RF in, IF out).
Both images appear: filter the RF side with r2 or r4.

r8 RX mixer, ADL5350

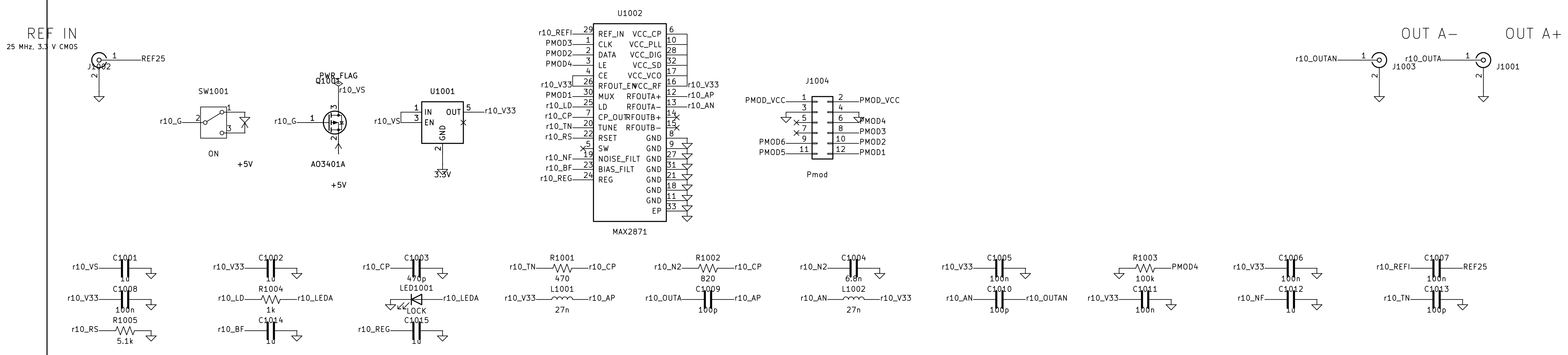
cells (2,2), (3,2)



EXPECT: RF x LO to IF (or IF x LO to RF) with 7−9 dB of loss from 400 MHz to 2.5 GHz and an IF of 0.1−30 MHz; +4 dBm of LO is enough
ADL5350: a GaAs FET switch mixer with its own LO buffer; the RF and IF pins are one node, split by a crossover (series 15 pF + shunt 47 nH toward RF, series 47 nH + shunt 15 pF toward IF, corner 200 MHz) in place of the datasheet's band−tuned tanks: a few dB more loss, every band at once. The LO's series element is 0 ohm (fit the datasheet's 3−8 nH for one band); the LO bias choke is 10 nH (the datasheet: 24 nH at 780 MHz, 2 nH at 2 GHz).
Passive, so the same tile up−converts a DAC tone (IF in, RF out) or down−converts an antenna's signal (RF in, IF out).
Both images appear: filter the RF side with r2 or r4.

r10 synthesizer, MAX2871

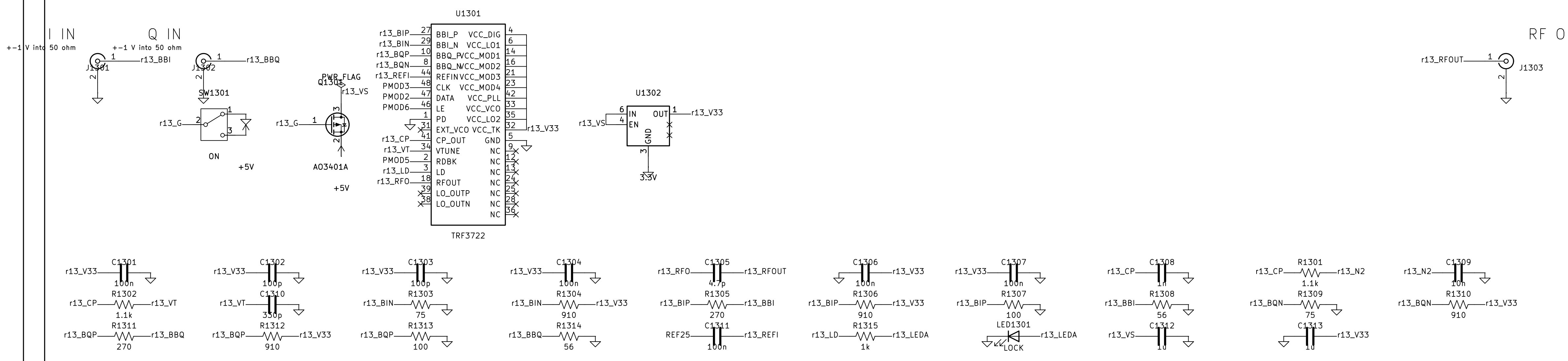
cells (0,3), (1,3)



EXPECT: any frequency from 23.5 MHz to 6 GHz on OUT A+ and OUT A− (each −4 to +5 dBm in steps, 180 degrees apart), locked to the 25 MHz on REF IN; the red LED says locked
MAX2871 as CostasLink built it: fractional−N PLL with four VCOs, programmed over SPI from the Pmod (SCLK, SDATA, LE; MUXOUT back). The loop filter is third order, 80−120 kHz with 61 degrees of margin for both 915 MHz (VCO / 4) and 2.4 GHz (VCO / 2) at ICP 2.56 mA (RSET 5.1 k, CP = 7). RFOUTA+ and RFOUTA− each feed a jack through 27 nH to 3.3 V and 100 pF: two outputs, 180 degrees apart; put a 50 ohm terminator on one that is unused. RFOUTB is open and must stay disabled. CE and RFOUT_EN are tied high, LE has 100 k down. REF IN is unterminated; a back trace takes it on to r13.

r13 I/Q transmitter, TRF3722

cells (2,3), (3,3)



EXPECT: I + jQ from two DACs on 0.4−4.2 GHz, about −3 dBm, image−free (−42 dBc untrimmed)
TRF3722 (PLL, VCO, I/Q modulator) on SPI from the Pmod (LE IO6, readback IO5); reference from r10's REF IN. Each DAC: 56 ohm at the jack, 270 ohm to a node held at 0.25 V (+−0.25 V for +−1 V). DC on I alone is a CW carrier.

GND GND GND GND

no mounting holes: four stick−on rubber feet, marked on the back

ground loops for the scope's clips, one per channel, in the board's margins

PWR_FLAG

Drawn by scripts/gen_sch.py from boards/rf_mod.py; the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
a quadrature transmitter, a synthesizer, three mixers (one a diode ring), an LNA and a gain block, two
Board: Jason Gallicchio and Claude | open hardware, CERN−OHL−P−2.0 | github.com/gallicchio/

Sheet: /
File: rf_mod.kicad_sch

Title: RF−MOD: 915 MHz, 2.4 GHz and a stretch to 5.8 GHz

Size: A1	Date: 2026−10−25	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1